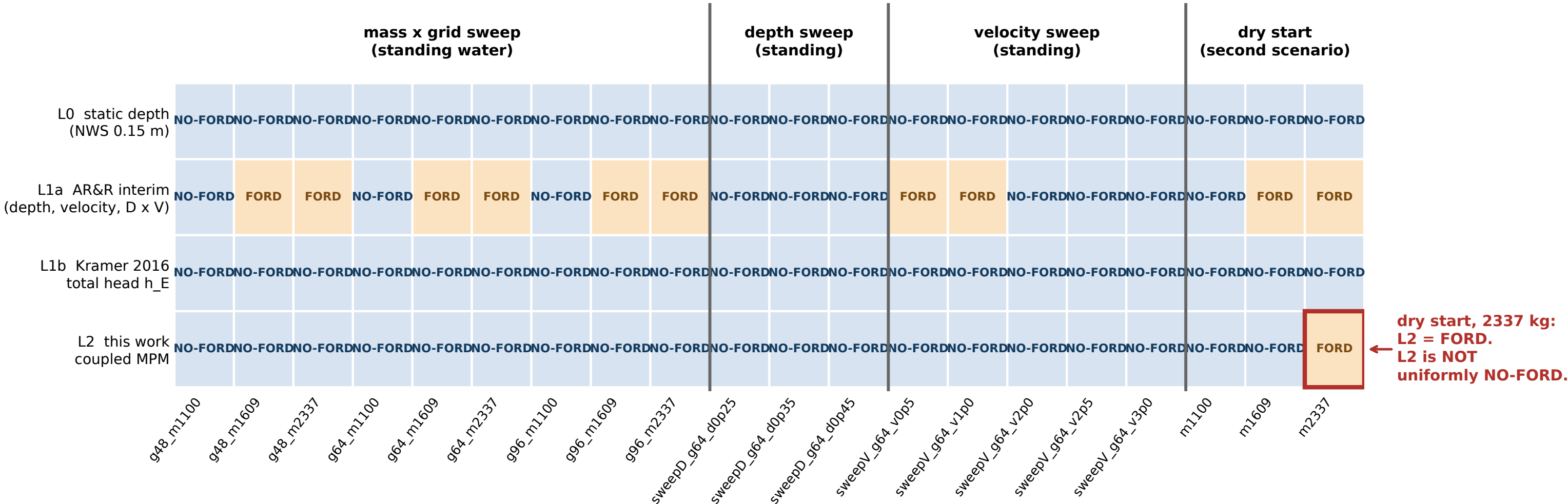


G3 Four-rung abstraction ladder, every run



Cells are read straight out of renders/yaris_render_s1/gates_results_all_runs.json, regenerated this session over all 20 runs. Nothing here is hand-typed.

L0 is the NWS Turn Around Don't Drown rule at 0.15 m, which is NOT AR&R. L1a is vehicle_params.L1_verdict, the full interim AR&R rule: depth cap AND velocity cap AND the D x V product cap, evaluated on realized depth. L1b is Kramer 2016 total head, $h_E = \text{depth} + v^2 / 2g = 0.409108 \text{ m}$ at the nominal operating point, tested against a 0.30 m passenger limit. L2 is this work, the coupled MPM simulation, NO-FORD when any frame exceeds 0.05 m of drift.

INDEPENDENCE CAVEAT: L0, L1a and L1b are all functions of (depth, velocity) alone. They share their entire input. Their agreement is one criterion evaluated three ways, NOT three independent confirmations. Only L2 reads the vehicle. L2 returns FORD in 1 of 20 runs, all in the dry-start scenario. The dry-start and standing-water runs are NOT a controlled pair: they come from different drivers (sim_dump.py vs sim_standing.py) and standing water re-injects inflow momentum every step, so the scenario difference cannot be attributed to the initial condition alone. Class names denote which AR&R limit set was applied by kerb weight, nothing more.